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RESEARCH ARTICLE

# Design of an Early Prediction Model for Parkinson's Disease Using Machine Learning

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**ABSTRACT** Parkinson's Disease (PD) is a chronic and progressive neurological disorder that impairs the body's nervous system pathways. This disruption results in multiple complications related to movement and control, manifesting as symptoms such as tremors, rigidity, and impaired coordination. In the initial phases of PD, individuals have trouble with speech and exhibit a slow rate of verbal expression. Dysphonia is seventy to ninety percent of persons with Parkinson's disease report a speech impairment or modification in speech, and it serves as a preliminary indicator of the disease. Consequently, speech can be a crucial modality in the initial phase of Parkinson's disease prediction. In literature, diverse Machine Learning models are employed for Parkinson's disease diagnosis utilizing voice analysis. Challenges such as class imbalance, feature selection, and interpretable predictive analysis still need to be addressed. Furthermore, the precision and reliability of the predictive outcomes are crucial to enhance healthcare services. Consequently, we propose an Explainable balanced Recursive Feature Importance with Logistic Regression (XRFILR) model to address the abovementioned issues. The proposed model extracts the pertinent features using an RFE with a Logistic Regression classifier and evaluates the feature significance in Parkinson's disease prediction using eXplainable Artificial Intelligence. We employed the seven machine learning classifiers the model offers to diagnose Parkinson's disease using significant speech data. Among these ML models, the proposed model achieved an accuracy of 96.46%, surpassing existing machine learning techniques.

**INDEX TERMS** Parkinson's disease, K-MeansSMOTE, recursive feature elimination, logistic regression classification with permutation, eXplainable artificial intelligence.

## I. INTRODUCTION

Parkinson's Disease (PD) is a neurodegenerative disorder that damages dopamine-producing cells in a particular region of the brain. PD is the second most prevalent neurological disorder following Alzheimer's Disease. According to a recent study between 2013 and 2023, the researchers introduced various computer-based approaches for PD prediction, including conventional medical imaging techniques such as Magnetic Resonance Imaging (MRI) and Single-Photon Emission Computed Tomography, as well as assessing speech patterns, handwriting movements, gait signals. Signals from electroencephalograms (EEGs) [1] electromyography (EMG) [2], and symptoms of Freezing of Gait (FoG) [3]. It is exceedingly challenging to detect in the initial phase. According to a current study, statistics from the World Health Organization indicate that over 8.5 million individuals worldwide had Parkinson's disease in 2019. Machine Learning (ML) has recently experienced a surge in prominence. It is widely utilized in multiple industries due to its ability for pattern detection [4], [5]. Self-assisted decision-support systems that use machine learning advancements [6], [7] facilitate enhanced judgment. Levodopa and Syndopa are among the pharmacological agents utilized in managing Parkinson's disease; the dosage of each medication is determined by the severity of the condition in [8]. Researchers use deep learning and machine learning techniques to enhance PD prediction. Seventy to ninety percent of those with Parkinson's disease have dysphonia or impaired speech. The initial warning indicator is disarray or alteration in speech. In 2021, the low tone, rapid bursts, strained voices, and prolonged pauses of Parkinson's disease patients rendered

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their speech difficult to understand. Parkinson's Disease affects males 1.5 times more frequently than females, with an estimated global population of 10 million individuals now diagnosed, and estimates suggest a rise in this number. Increase to 1.2 million by 2023. Each year, some 60,000 persons in the United States are diagnosed with Parkinson's disease, predominantly affecting adults over the age of 50. recently discovered a 4% prediction rate of Parkinson's disease cases in persons. Reference [7] Saravanan et al., 2022 [9] the early detection of PD possesses substantial scientific importance. Due to its advanced attributes and ability to enhance patient outcomes and quality of life. Moreover, early diagnosis may enable the prediction of the emergence of additional neurodegenerative disorders, given the similarities in their symptoms. Artificial Intelligence (AI) techniques, particularly Machine Learning (ML), have received considerable focus in medical diagnostics owing to their ability to examine large datasets and generate accurate predictions. Shastry, 2023 [10] used Sampling strategies such as oversampling and undersampling to balance datasets, improve performance, and reduce bias towards the majority class, so class distribution is important in predicting PD using ML techniques. Ali et al. 2023 [11] identified that computing efficiency for high-dimensional dataset complexity requires proficient feature selection. They aimed to tackle computational issues using an Ensemble Feature Selection Algorithm (EFSA). Bukhari and Ogudo 2024 [12] introduced an ensemble machine learning model that uses the AdaBoost classifier to detect PD from speech signals, utilizing SMOTE for data balancing and PCA for dimensionality reduction. As a non-invasive, cost-effective diagnostic tool, it achieves exceptional performance (96% accuracy, 98% precision, and AUC 0.99). However, the model's computational complexity, the necessity for hyperparameter adjustment, and reliance on a limited dataset challenge broader clinical use, excluding the limitations of the explainable AI aspect of this work. Shyamala and Navamani 2024 [13] proposed an Interpretable Feature Ranking XGBoost (IFRX) model for predicting PD using speech data and performed better. The authors have done preprocessing, SVMSMOTE for class balancing, RFE with XGBoost for feature selection, and SHAP for interpretability but used a small dataset size. Notwithstanding substantial medical progress, Parkinson's disease remains a global issue. Current estimates suggest that Parkinson's disease resulted in approximately 329,000 fatalities in 2019. Since 2000, it has increased by over 100 percent by 2024 [14]. The early diagnosis of Parkinson's disease is vital for disease prediction and necessitates a method that identifies critical features for predicting Parkinson's disease. In literature, ML techniques are employed for PD prediction. Nonetheless, challenges such as class imbalance, feature selection, and explainable artificial intelligence are not efficiently managed. We intended to create an efficient predictive model for the early diagnosis of Parkinson's Disease utilizing the speech dataset. The primary novel methodology for the proposed Explainable balanced Recursive Feature Importance with Logistic Regression (XRFILR) model seeks to optimize feature identification and improve classification accuracy.

The significant contributions of this work are as follows:

1. Proposed the Explainable balanced Recursive Feature Importance with Logistic Regression (XRFILR) model, which utilizes speech features exhibiting superior stability, strength, and efficacy in achieving optimal classification performance relative to alternative machine learning methods.

2. Enhanced the accuracy of Parkinson's disease predictions by employing the KMeans Synthetic Minority Oversampling Technique (KMeansSMOTE) to balance the extensive speech dataset.

3. Enhanced model interpretability and identified the most significant elements for accurate prediction of PD using SHapley Additive exPlanations (SHAP) Analysis.

4. A comparative analysis was also made with the state of the art of the research works with the proposed model.

The study's structure is organized as follows: Section II discussed the related works of Parkinson's disease. The Proposed model delineates the workflow, incorporating feature selection, classification, and SHAP-based interpretability, presented in section III. Experimental results demonstrate that XRFILR outperforms alternative approaches, and the discussion examines findings and model comparisons in section IV. The Conclusion encapsulates contributions and proposed avenues for future works in section V.

## II. RELATED WORKS

A multitude of research has been undertaken to predict Parkinson's Disease (PD) in patients. PD is the second most common neurodegenerative disorder that impacts speech and vocal precision. The onset of PD transpires between the ages of 50 and 65. Many studies have investigated the application of speech signals for diagnosing PD, utilizing diverse features and approaches. This study examines current developments in employing the Explainable balanced Recursive Feature Importance with Logistic Regression (XRFILR) model to tackle the difficulties of identifying Parkinson's disease via speech analysis. It offers innovative perspectives on improving categorization precision and comprehensibility. Machine learning techniques have demonstrated significant utility in analyzing speech data, facilitating the extraction of pertinent features for predicting and monitoring Parkinson's disease.

Wang et al. 2020 [15] presented a deep-learning model for the early diagnosis of Parkinson's disease (PD), including premotor signs, including Rapid Eye Movement (REM) anomalies, olfactory impairment, cerebrospinal fluid biomarkers, and dopaminergic imaging markers. The model surpasses twelve other machine learning techniques, with an average accuracy of 96.45% on a dataset including 183 healthy persons and 401 early-stage PD patients. Furthermore, it offers insights into feature significance derived from the Boosting approach, improving model interpretability. There is a deficiency in verifying the model on more extensive datasets, integrating more data types, and investigating real-time monitoring for ongoing early detection. Hoq et al. 2021 [16] integrated two models and derived from a Support Vector Machine (SVM) integration using Principal Component Analysis (PCA) with a Sparse Autoencoder (SAE) for the detection of PD patients based on their vocal features. The PD speech dataset was equilibrated according to the SMOTE technique, resulting in an accuracy of 94.4%. PCA and SAE are employed in the research to select whether there is an inadequacy in the efficacy of identifying the essential features for diagnosing Parkinson's disease. Soumaya et al. 2021 [17] suggested that advancing a precise Parkinson's disease prognostication system is significant. The authors employed an enhanced methodology and assessed the optimization methodology of evolutionary algorithms together with speech acoustic and decomposition properties for the Support Vector Machine classifier. Liu et al. 2021 [18] proposed a comprehensive ensemble methodology. PD prediction is emphasized in Local discriminant Preservation Projection (LPP). The ensemble technique maintains the neighborhood configuration of PD speech samples while simultaneously augmenting the inter-class variance and diminishing the intra-class variance of the samples using an optimal approach. Unbalanced PD samples were adequately managed. Feature Selection ought to have been employed to mitigate the curse of dimensionality, which may also reduce dimensionality.

Goyal et al. 2021 [19] evaluated various classification methodologies to illustrate the efficacy of each classifier. To reduce the cardinality of the speech dataset without substantially undermining precision by looking at three separate factors, the Significance of Genetic Algorithm (GA) techniques, maximum Relevance Minimum Redundancy (mRMR), and Principal Component Analysis (PCA) were also examined. Two categories of datasets were utilized. Enhanced precision was achieved with The XGBoost classifier independent of class imbalance. The utilization of vocal features in diagnostic processes of Parkinson's disease is examined for the potential use of feature selection techniques.

Dhar 2022 [20] enhanced the categorization of advanced Parkinson's disease using a distinctive feature set (40PD, 40 Healthy Control) and the MI-AE-GOLGBM methodology, achieved a performance rate of 84.17%. Future initiatives concentrated on incorporating methods to enhance classification efficacy. Liu et al. 2022 [21] conducted a study on the remote monitoring of patients, employing a feature set comprising 50 attributes. This study employed Shapley Additive explanations with a Light Gradient Boosting Machine, SHAP-LightGBM, demonstrating a novel feature selection and modeling approach. The proposed model achieved a performance outcome of 91.62%. Traditional feature selection technique used. Lamba et al. 2022 [22] evaluated the effectiveness of Naive Bayes, k-NN, and Random Forest (RF) algorithms in diagnosing Parkinson's disease. The proposed method incorporated a Hybrid Mutual Information Ranking and Feature Elimination MIRFE-XGBoost, showcasing a unified feature extraction and selection strategy. The performance outcomes ranged from 95.58% to 95.47%, but lack of transparency in predicting the PD.

Hawi et al. 2022 [23] evaluated the efficacy of intense voice therapy in improving functional communication. Numerous researchers have made significant efforts to diagnose Parkinson's disease. Alalayah et al. 2023 [24] introduced an innovative approach to enhancing the tactics for early detection of Parkinson's Disease through speech analysis. Biases may arise if features are unintentionally omitted, excluded, or disproportionately represented. The model proficiently identified the symptoms of PD, exhibiting elevated classification accuracy.

Iyer et al., 2023 [25] utilized a pre-trained CNN (Inception V3) with transfer learning to analyze spectrograms derived from voice recordings of individuals diagnosed with Parkinson's. The deep learning model enhanced performance in identifying patients with Parkinson's disease, with an AUC of 0.97 for colored spectrograms and 0.96 for Monochromatic spectrograms. Rahman et al., 2023 [26] aim to diagnose Parkinson's disease in its early stage by employing various machine learning methods and a three-layer Deep Neural Network2 (DNN2), which surpasses all other methods with an accuracy of 95.41%. This research indicated that DNNs demonstrate enhanced efficacy relative to conventional machine learning techniques in patient categorization utilizing PD derived from voice signals. Alshammri et al., 2023 [27] assessed many ML models to build a soft diagnostic tool for identifying Parkinson's disease using vocal signal features. The research revealed that the model based on Support Vector Machines (SVM) exceeded previous machine learning models in diagnosing patients with PD, achieving a remarkable accuracy of 96%. Saxena and Andrew 2023 [28] used machine learning models such as SVM, KNN, Random Forest, and Logistic Regression to develop a classification framework for Parkinson's disease detection using voice input. It demonstrates Random Forest's efficacy by achieving the highest accuracy of 89.47% on the testing set. However, the proposed model does not address class imbalance or explainable insights, which are critical for improving the interpretability and trust of medical diagnosis.

Chawla et al. 2024 [29] introduced a method utilizing the Zebra Optimization Algorithm (ZOA) and Recursive Feature Elimination Cross-Validation (RFECV) in the context of Nature-Inspired approaches. This methodology utilizes feature selection Novel Information Filter System (NIFS) to ascertain PD. It offers the potential for diagnosing more disorders, contingent upon the dataset's numerous properties. Singh and Tripathi 2024 [30] used voting techniques to determine the final prediction, enhancing the reliability of diagnosis in early Parkinson's disease detection. A notable limitation of this approach is its reliance on computational resources due to the complexity of simultaneously training multiple models. Additionally, The model's accuracy may still be contingent upon the training dataset's quality and size and not using explainable AI.

Bukhari and Ogudo 2024 [12] introduced an ensemble machine learning model that uses the adaBoost classifier to detect PD from speech signals, utilizing SMOTE for data balancing and PCA for dimensionality reduction. As a non-invasive, cost-effective diagnostic tool, it achieves exceptional performance (96% accuracy, 98% precision, and AUC 0.99). However, the computational complexity of the model, the necessity for hyperparameter adjustment, and reliance on a limited dataset challenge broader clinical use, excluding the limitations of the explainable AI aspect of this work. Shyamala and Navamani 2024 [13] proposed a model called Interpretable Feature Ranking XGBoost (IFRX), which has a sequence of processes such as SVMSmote for class balancing, RFE with XGBoost for feature selection with classifier and Shap for explainability. Achieved better performance but the authors used a small dataset for their work. Hossain and Amenta 2024 [31] used pipelines, offering higher classification metrics (85.09% accuracy, 92% precision). Hence, this study stresses the potential of machine learning classifiers and pipelines for distinguishing Parkinson's disease patients from healthy controls using speech biomarkers. The pipelining technique successfully addresses the significance of features in high-dimensional data, improving classification accuracy. As a result, the interpretability and therapeutic efficacy of the study are limited because it fails to address class imbalance or employ explainable artificial intelligence techniques.

The above literature review indicates that several studies have not adequately addressed the problem of class imbalance, potentially resulting in biased model performance that favors the majority class, thereby undermining the trustworthiness of the diagnosis. An increase in features leads to decreased model accuracy due to numerous irrelevant and linked feature subsets. Consequently, the dataset must be refined to incorporate the most pertinent features. Existing research indicates that feature selection techniques are crucial for identifying the most relevant features and enhancing the accuracy of various machine learning classification algorithms. Numerous recent research underscores the advantages of using explainability approaches, such as Shapley Additive Explanations (SHAP), to elucidate model predictions and offer insight into the role of certain vocal features in PD classification. Moreover, ensemble learning and hybrid methodologies have demonstrated efficacy in mitigating class imbalance, a prevalent issue in medical datasets, and enhancing feature selection to prevent duplication and overfitting.

Here, we propose an Explainable Recursive Feature Importance with Logistic Regression (XRFILR) model, which equilibrates class distribution and emphasizes selecting the most informative features. The XRFILR model seeks to improve interpretability and precision in predicting PD based on voice features, contributing to a more sophisticated and dependable diagnostic instrument. This technique aims to enhance the shortcomings of existing studies and progress the domain of speech-based PD diagnosis by utilizing balanced datasets, recursive feature selection with logistic regression, and interpretability.

**TABLE 1. Essential speech features used for parkinson's disease prediction.**

| Feature | Description | Type | Units /Range |
|---|---|---|---|
| MDVP (Multi-Dimensional Voice Program) | Average vocal fundamental frequency | Numeric | Hz |
| MDVP | Maximum vocal fundamental frequency | Numeric | Hz |
| MDVP | Minimum vocal fundamental frequency | Numeric | Hz |
| Jitter (%) | Frequency variation in the voice | Numeric | % |
| Jitter (Abs) | Absolute frequency perturbation | Numeric | ms |
| Shimmer | Amplitude variation in the voice | Numeric | dB |
| Shimmer (dB) | Logarithmic amplitude perturbation | Numeric | dB |
| HNR (Harmonic-to-noise ratio) | Harmonic-to-noise ratio | Numeric | dB |
| RPDE (Recurrence period density entropy) | Recurrence period density entropy (measures the randomness of speech) | Numeric | [0, 1] |
| DFA (Detrended fluctuation analysis) | Detrended fluctuation analysis (correlates with speech signal variation) | Numeric | [0, 2] |
| PPE (Pitch period entropy) | Pitch period entropy (measures pitch irregularity) | Numeric | [0, 1] |
| NHR (Noise-to-harmonic ratio) | Noise-to-harmonic ratio (a measure of vocal noise) | Numeric | Ratio |
| Spread1 | Non-linear vocal fold structure measurement | Numeric | Dimensionless |
| Spread2 | Second non-linear vocal fold structure measurement | Numeric | Dimensionless |
| D2 | Correlation dimension (measures complexity of vocal signal) | Numeric | Dimensionless |
| Class | PD diagnosis (0: Healthy, 1: PD) | Binary | 0, 1 |

## III. PROPOSED EXPLAINABLE BALANCED RECURSIVE FEATURE IMPORTANCE WITH LOGISTIC REGRESSION (XRFILR) MODEL

The proposed XRFILR model encompasses procedures including Data Preparation, Exploratory Data Analysis, Class Balancing, Feature Selection, Classification, Hyperparameter Tuning, Model Training, and Explainable Artificial Intelligence.

*Dataset Description:*

University of California Irvine (UCI) Machine Learning Repository was used to evaluate our proposed model for Parkinson's disease and benchmark Dataset [32]. This dataset has speech-related features extracted, including Parkinson's disease (PD) patients and Healthy Controls (HC). The dataset demonstrates an imbalance, consisting of 188 patients with Parkinson's disease (107 men and 81 females) and 64 healthy controls (23 males and 41 females). The primary aim of this dataset is classification, designed to distinguish between PD and HC patients based on complex voice patterns. Table 1 delineates essential features frequently employed in PD prediction, particularly utilizing voice data. These features originate from several facets of voice production and signal processing, encompassing frequency and

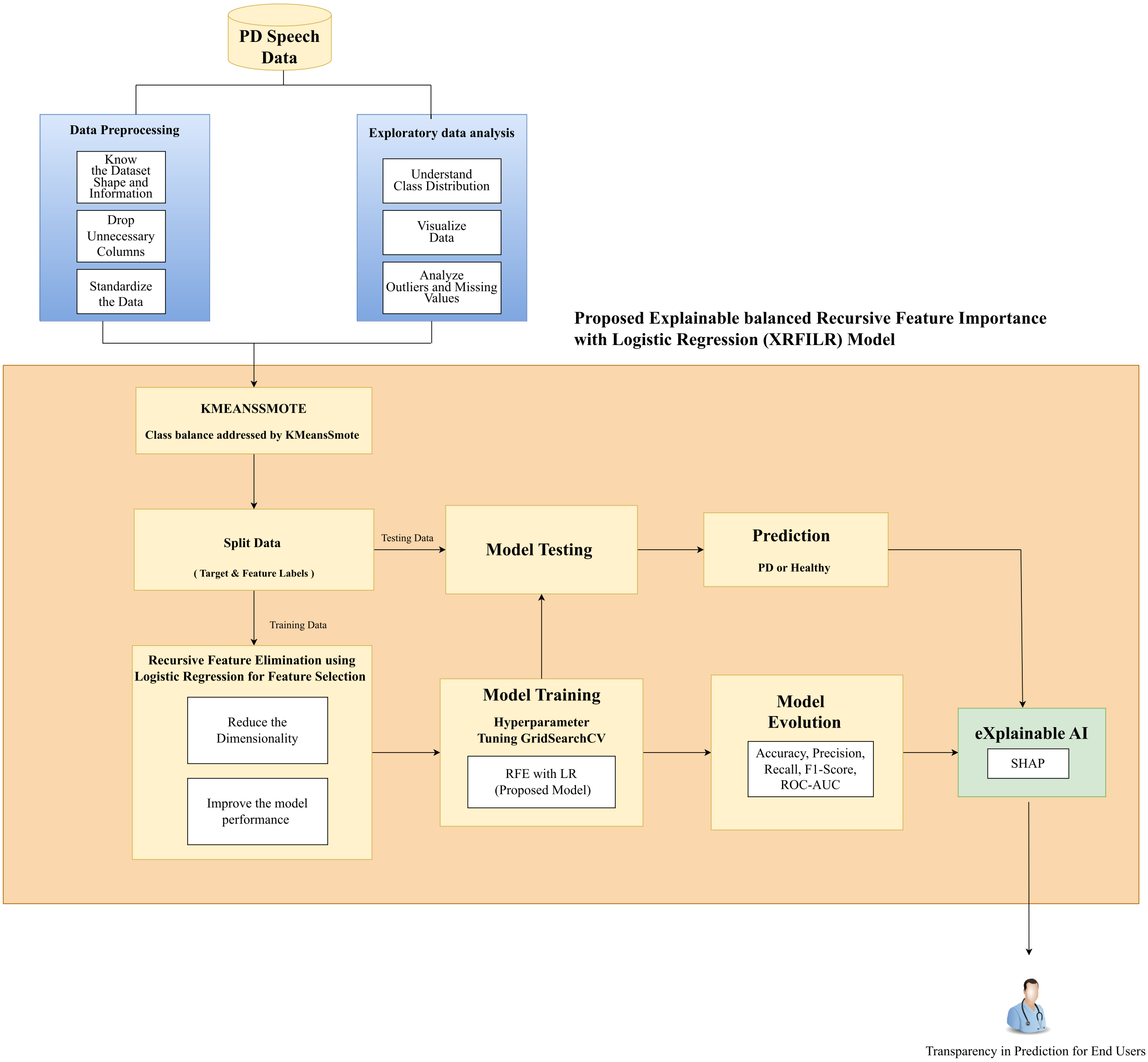



**FIGURE 1. Architectural diagram of the proposed system.**

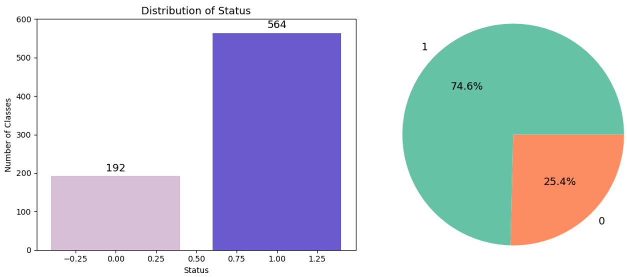
amplitude fluctuations alongside speech signals' intricacy and stochastic nature. Collectively, these features establish a comprehensive model for detecting and monitoring PD using speech analysis, with each attribute reflecting distinct areas of vocal performance often impaired in persons with PD.

## A. PROPOSED MODEL

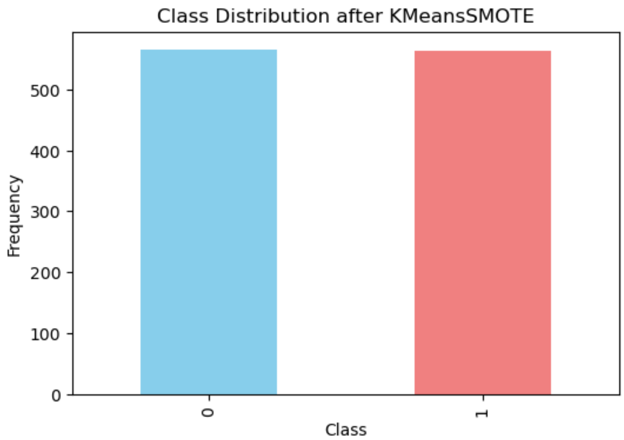
Figure 1 illustrates the overall design of the proposed system and starts with the data collection and preprocessing phases, which are crucial in predicting Parkinson's disease and encompass target variable identification, standardization, validation, and oversampling. Feature extraction is performed to enhance the model by finding essential elements and eliminating irrelevant ones. A classification algorithm pipeline is established, and model assessment is performed via cross-validation. SHAP is employed to demonstrate the importance of global features. The ML process classifies Parkinson's Disease utilizing a speech dataset. The procedure encompasses data preprocessing, class balancing, feature extraction, model training, and assessment utilizing accuracy, precision, recall, F1-score, and ROC-AUC metrics. The outcome categorizes whether a particular speech sample is associated with a PD patient or a healthy control.

## B. DATA PREPROCESSING

The process begins with the importation of the dataset as numeric data. The dataset has been loaded and examined to comprehend its structure and features. The data preparation




**(a)** Class Imbalance Representation




**(b)** Class Balancing after KMeansSMOTE

**FIGURE 2.** **Class balancing using KMeansSmote.**

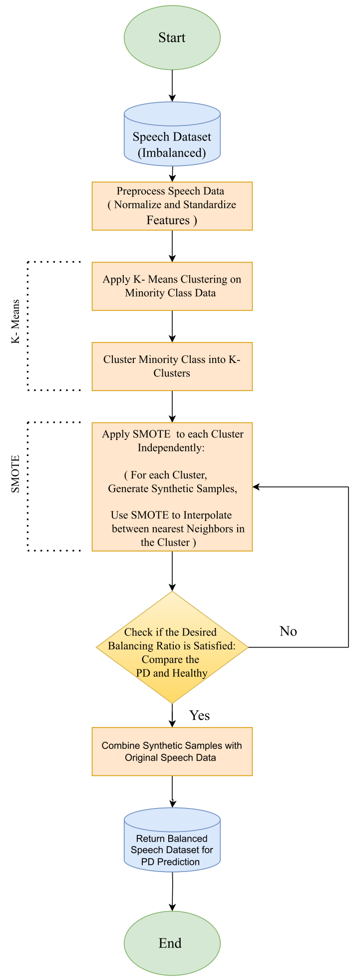



**FIGURE 3.** **Flowchart of KMeansSMOTE technique.**

phase entails verifying the dataset's structure and details. The preprocessing pipeline for this study is designed to enhance classification performance by preparing the data for effective ML modeling. First, the dataset undergoes an initial cleaning phase where irrelevant columns, such as 'id' and 'class,' are dropped. The features are then standardized using the Standard Scaler, which ensures that each feature has a mean of zero and a standard deviation of one. This step is critical for models sensitive to feature scaling, such as Logistic Regression. Exploratory Data Analysis aims to grasp the data comprehensively and the interrelationships among variables, facilitating the selection of the most pertinent features for model training. A correlation heatmap assesses the relationships among different attributes. Standardization ensures that all features are on a uniform scale, with numerical features adjusted to have a mean of 0 and a standard deviation of 1. This procedure enhances overall data quality, augments model precision, and eradicates data bias. The procedure started with analyzing the distributions and correlations of individual attributes. Subsequently, the data was normalized and balanced to address class imbalance issues.

### C. RESAMPLING USING KMEANSSMOTE

The KMeansSMOTE technique is applied to address the class imbalance problem. KMeansSMOTE combines synthetic minority oversampling with clustering to generate new samples that reflect the intrinsic structure of the minority class while maintaining the distribution of the data, as shown in Figures 2a and 2b. Feature extraction is a crucial aspect of preprocessing. It diminishes data dimensionality, enhances model performance through relevant data, and allows for a more precise representation of underlying patterns. It is crucial in balancing class tasks as it transforms the original feature data into a new collection of more distinct and informative features. This transition can mitigate the effects of class imbalance by augmenting the uniqueness of the minority class to the dominant class.

KMeansSMOTE (Synthetic Minority Over-sampling Technique) is an effective method to rectify class imbalance in datasets, especially in predicting Parkinson's disease (PD) through speech data. This technique elucidates using KMeansSMOTE on a speech dataset for PD prediction. KMeansSMOTE integrates the KMeans clustering algorithm with the SMOTE technique to produce synthetic samples for the minority class, as shown in Figure 2. This is particularly advantageous when the minority class is inadequately represented relative to the majority class. In Parkinson's disease prediction, the dataset generally has two categories: one denoting individuals with Parkinson's disease and the other denoting healthy persons. The initial phase involves identifying the minority class, typically the PD class, alongside the majority class. Subsequently, KMeans clustering is executed on the minority class samples.

KMeansSMOTE entails choosing a designated number of clusters (k) and implementing the KMeans algorithm on the minority samples. The clustering procedure categorizes the minority class occurrences into k clusters from the speech dataset. KMeansSMOTE produces synthetic samples for every cluster created within the minority class. The procedure entails the subsequent steps: choosing a random sample from the cluster, determining its k-nearest neighbors within the same cluster, and producing synthetic samples using interpolation between the selected sample and its neighbors. It generates a broader array of synthetic samples by considering the data's local structure, enhancing the quality of the produced examples. This helps improve model generalization and ensures a more balanced classification, as shown in Figure 3. After this resampling step, Recursive Feature Elimination (RFE) reduces the feature set to the most informative 50 features, further enhancing model efficiency and performance. The data is divided into training and testing sets to assess model performance in real-world scenarios.

### D. DIMENSIONALITY REDUCTION

This study emphasizes feature extraction as a crucial process to enhance the model's performance and interpretability by identifying the most pertinent features from the high-dimensional dataset. The procedure commences with Recursive Feature Elimination (RFE), a methodical technique for identifying a subset of the most significant features and top ten features listed in Table 2 based on feature selection.

#### 1) RECURSIVE FEATURE ELIMINATION (RFE) WITH LOGISTIC REGRESSION CLASSIFIER

Figure 4 shows the RFE flow chart with logistic regression. The RFE approach operates by repeatedly training a Logistic Regression model, assessing the significance of the features, and discarding the least relevant ones. The algorithm identifies the 50 most informative features from the extensive feature collection. It ensures that only the model's most predictive attributes are provided, decreasing dimensionality and enhancing computational efficiency. This study uniquely integrates KMeansSMOTE to tackle class imbalance and RFE for feature selection simultaneously, improving model performance for imbalanced datasets on Parkinson's disease classification. This combination ensures that synthetic samples from the minority class preserve the most pertinent features, thus enhancing prediction accuracy and mitigating the danger of overfitting. Applying permutation importance to the optimal model enhances feature significance, providing an extra dimension of interpretability, as shown in Figure 5. The RFE approach operates by repeatedly training a Logistic Regression model, assessing the significance of features, and discarding the least pertinent ones. The algorithm identifies the 50 most informative features from the extensive feature collection. It ensures that only the model's most predictive attributes are provided, decreasing dimensionality and enhancing computational efficiency. This study uniquely integrates KMeansSMOTE and RFE, improving model performance for imbalanced datasets on Parkinson's disease classification.

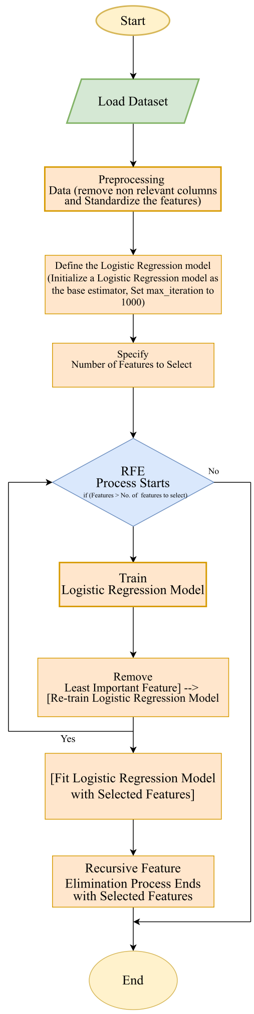



**FIGURE 4.** Recursive feature elimination with logistic regression.

Table 2 presents the ten most significant aspects for predicting Parkinson's disease utilizing a speech dataset comprising 756 features. An index denotes each feature, and its corresponding significance value indicates its influence on the model's predictive accuracy. Elevated significance values signify features that play a more substantial role in the model's decision-making process. This ranking offers insights into the most relevant speech qualities associated with PD.

As shown in Figure 5, the Feature Importance (Permutation) plot for Logistic Regression was utilized to predict Parkinson's disease (PD) based on speech data. In this context, the importance of the permutation feature quantifies the decline in model performance when the values of a given feature are randomly shuffled, revealing the model's reliance on that feature for prediction. A more excellent significance

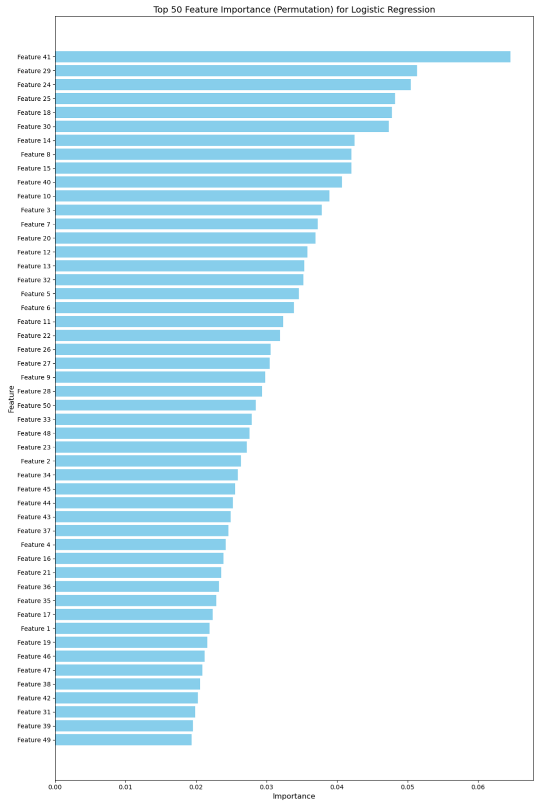



**FIGURE 5. Permutation feature importance for logistic regression model.**

score indicates that the features exert a more substantial influence on the classification model.

Feature 41 represents a non-linear voice metric from PD speech datasets, such as Harmonics-to-Noise Ratio (HNR) or Shimmer. These features reflect fluctuations in the voice, specifically regarding the uniformity and steadiness of pitch and loudness. Individuals with Parkinson's disease frequently demonstrate vocal cord dysfunction, resulting in alterations such as hoarseness, monotonic speech, or breathiness. HNR and analogous metrics evaluate the extent to which a patient's vocal signal is predominated by harmonic content instead of noise, elucidating subtle motor symptoms that influence speech output. Feature importance elucidates the speech-based elements that most significantly influence the model's decisions, enhancing model interpretability. These insights are essential for formulating effective Explainable AI (XAI) techniques in clinical environments, ensuring precise predictions and confidence in the model's results.

Recursive Feature Elimination (RFE) is a backward feature selection method employed to discern and preserve the most crucial features by systematically removing less significant ones. During this procedure, an estimator (Logistic Regression) is trained on the dataset, and the significance of features is assessed using the model's coefficients or relevance scores. The least significant features are systematically eliminated until the target number of features is attained. This study employs Recursive Feature Elimination (RFE) with a

**TABLE 2. Top 10 feature index and importance values.**

| Feature Index Value | Feature Importance Value |
|---|---|
| 41 | 0.064602 |
| 29 | 0.051327 |
| 24 | 0.050442 |
| 25 | 0.048230 |
| 18 | 0.047788 |
| 30 | 0.047345 |
| 14 | 0.042478 |
| 8 | 0.042035 |
| 15 | 0.042035 |
| 40 | 0.040708 |

Logistic Regression classifier as the foundational estimator, as shown in Figure 4. Logistic Regression uses the logistic function to evaluate the correlation between input features and a binary target.

Figure 4 shows the workflow of the RFE with logistic regression. A Logistic Regression model is established as the base estimator, with a maximum iteration limit of 1000 to ensure model convergence. The quantity of features to preserve is determined, frequently relying on past information or heuristics. The RFE process commences, during which the Logistic Regression model is trained on the complete feature set, and features are recursively assessed for their significance. In each iteration, the least significant feature is chosen and eliminated, after which the model is retrained using the remaining features.

This procedure persists till the target quantity of features is attained. After selecting the ideal subset of features, a final Logistic Regression model is constructed using the selected features. The RFE process culminates with the final model prepared for assessment or implementation. This approach ensures the retention of the most pertinent features, enhancing the model's interpretability and precision in predicting PD.

### 2) TIME COMPLEXITY

In investigating the time complexity of the Parkinson's Disease (PD) prediction model, we concentrate on specific critical processes that influence the total computational expense. The overall time complexity is determined by evaluating the several processes involved, including preprocessing, feature selection, and model interpretation, as shown below:

1) Preprocessing (StandardScaler): $O(n \cdot d)$
2) KMeansSMOTE: $O(k \cdot n \cdot t)$
3) Recursive Feature Elimination (RFE): $O(F \cdot n \cdot d^2)$
4) Logistic Regression: $O(n \cdot d)$
5) Permutation Feature Importance: $O(F \cdot n \cdot d)$
6) SHAP for Logistic Regression: $O(n^2 \cdot d)$

where:

- $n$: number of samples.
- $d$: number of features.
- $k$: number of clusters for KMeans in SMOTE.
- $t$: number of iterations in KMeans.

- $F$: number of features selected/eliminated by RFE.

**Total Time Complexity:**
$O(\max(F \cdot n \cdot d^2, n^2 \cdot d))$

### E. MODEL TRAINING AND EVALUATION

During the model training and evaluation phase, various machine learning classifiers are utilized on the preprocessed and balanced dataset to classify occurrences of Parkinson's disease. The employed classifiers comprise Logistic Regression [33], Random Forest [34], Support Vector Machine(SVM) [35],Multi-layer Perceptron (MLP) [36], CatBoost [37], XGBoost [38], and K-Nearest Neighbors (KNN) [39]. Each model is trained using features selected by Recursive Feature Elimination (RFE) with logistic regression and assessed on a test set. The training set, including 80% of the data, is utilized for model training. The most significant features were selected to enhance data quality and the model's overall performance. After assessing each feature's value, the other features were eliminated from the dataset. We utilize a series of models to select the most suitable one for training, ensuring great precision in generating accurate predictions and risk percentages. The proposed model addresses class imbalance when one class has significantly fewer samples than another in a binary classification context. In such circumstances, a model's performance may be suboptimal as it will demonstrate a bias towards the majority class.

Using a training and testing set to split, the dataset is split into 80% training and 20% testing. Each model is trained using the training set and evaluated on the test set. Performance indicators, including accuracy, precision, recall, and F1-score, are computed to evaluate each model's generalization capability on unseen data. Moreover, Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) are employed to evaluate models based on their categorization criteria. Confusion Matrix is created to comprehensively analyze true positive, true negative, false positive, and false negative rates. The values of various relevant matrices determine the efficacy of a classification model. The confusion matrix is for performance measurement and assists in reviewing and evaluating the model's efficacy. These facilitate the comparison of alternative categorization models and the identification of the optimal model for each parameter. The confusion matrix, recall, precision, F1-score, and accuracy are often employed as performance metrics [40]. The Confusion Matrix is an effective tool for evaluating the accuracy of a model. Statistical measures of accuracy, precision, recall, and F1 score were employed to assess the efficacy of classification algorithms applied to the PD voice dataset. The methodology for calculating classified instances is delineated in equations (1) to (4), where accurately categorized cases are denoted as TP (True Positive) and TN (True Negative). Conversely, misclassified occurrences are denoted by FP (False Positive) and FN (False Negative).

$$\text{Accuracy} = \frac{TN + TP}{TN + TP + FN + FP} \tag{1}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

$$F1\ \text{score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

The mathematical model delineates the procedures for managing class balancing, the feature selection technique, and the training and interpretability aspects. The result is a logistic regression model that predicts the existence of PD with interpretable feature significance, allowing for the quantification and elucidation of each feature's contribution to the prediction shown in algorithm 1.

### F. EXPLAINABLE AI

Explainable Artificial Intelligence (XAI) methodologies such as SHAP (SHapley Additive exPlanations) and Local Interpretable Model-Agnostic Explanations (LIME) elucidate the predictions of machine learning models. LIME augments interpretability by offering localized explanations for the predictions generated by the models. LIME facilitates the identification of the factors that most significantly influence the predictions of specific cases, providing clarity into the model's decision-making process on a case-by-case basis. SHAP is based on a notion from game theory known as Shapley values. These values facilitate the allocation of credit (or blame) among participants in a collaborative game. In machine learning, the "players" refer to the many characteristics utilized by the model, while the "game" denotes the prediction process. SHAP denotes SHapley Additive exPlanations. This approach elucidates the internal mechanisms of any machine learning model, namely its predictive processes. Cooperative game theory offers a methodology for computing the Shapley value of the model. The SHAP (Shapley Additive explanations) approach is utilized to improve the interpretability of machine learning models for categorizing Parkinson's disease. SHAP quantifies the contributions of specific features to the model's predictions using Shapley values, which are based on cooperative game theory. Tree-based models such as Random Forest, XGBoost, and CatBoost and utilize the Tree Explainer to calculate SHAP values, clearly illustrating the impact of each feature on the model's output for the positive class [13].

The Kernel Explainer is utilized for non-tree models, including Logistic Regression, SVM, MLP, and KNN, providing a model-agnostic method to compute feature contributions. The outcomes are illustrated using SHAP summary graphs, which demonstrate the significance of each characteristic overall test instance, emphasizing their average impact on the model's predictions. This method not only identifies the most influential features but also elucidates the underlying decision-making processes of the models,

**Algorithm 1** Mathematical Analysis of Interpretable Logistic Regression Model for Parkinson's Disease Diagnosis

**Input dataset:** $D = \{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$, where $x_i \in \mathbb{R}^d$ represents the feature vector and $y_i \in \{0, 1\}$ is the label (0 for no Parkinson's Disease, 1 for Parkinson's Disease)
**Target variable:** $y \in \{0, 1\}^n$, a vector representing the PD status for each sample.
**Preprocessing:**
Standardize the feature matrix: $x'_{ij} = \frac{x_{ij}-\mu_j}{\sigma_j}$, where $\mu_j$ and $\sigma_j$ are the mean and standard deviation of feature $x_j$.
**Class imbalance addressing:** Apply KMeans-SMOTE to generate synthetic minority class samples.
Apply K-Means clustering to the minority class samples to obtain centroids: $c_j = \frac{1}{|S_j|} \sum_{x \in S_j} x$, for $j \in \{1, 2, \ldots, k\}$.
Generate synthetic samples along the lines connecting centroids and their nearest neighbors: $D' = \{(x'_1, y'_1), (x'_2, y'_2), \ldots, (x'_{n'}, y'_{n'})\}$.
**Feature Selection with Recursive Feature Elimination (RFE):**
Define classifier: $h(x) \in \{0, 1\}$, a binary classifier (Logistic Regression).
Initialize: Set $S = \{1, 2, \ldots, d\}$ (all features).
Set $k =$ desired number of features to select.
**while** $|S| > k$ **do**
Train Logistic Regression model: $h_S = \text{train}(LogReg, D_S, y)$ on features in $S$ where $h_S$ is the trained model and $D_S$ is the dataset with selected features.
Compute feature importance: $\phi_i = \text{feature\_importance}(h_S, i)$ for $i \in S$.
Remove feature with lowest importance: $S = S \setminus \{\arg\min_i \phi_i\}$.
**end while**
**Output:** Selected features $S$.
**Model Training:**
**for** $k = 1$ to $K$ **do**
Train classifier: $h_k = \text{train}(M_k, D_{\text{resampled}})$, where $h_k$ represents the classifier trained using machine learning algorithms $M_k$ (Logistic Regression).
**end for**
**Model evaluation:**
Use cross-validation to evaluate model performance on unseen data.
Compute performance metrics: Accuracy, AUC, Precision, Recall, F1-score.
**Model explainability:**
Compute SHAP values for interpretability: $\Phi = \text{SHAP}(h, D_{\text{resampled}}.X)$, SHAP values calculated for the trained model on the resampled dataset.
**Output:** Contribution of individual features to model prediction and transparency for decision-making.

thereby enhancing transparency and reliability in predicting Parkinson's disease. The XRFILR model markedly diverges from current models by including KMeansSMOTE and Recursive Feature Elimination (RFE) to tackle critical issues in the early prediction of Parkinson's disease. In contrast to conventional models that use fundamental oversampling methods such as SMOTE, KMeansSMOTE in XRFILR facilitates targeted oversampling by grouping minority class samples and producing synthetic data within these clusters, minimizing overfitting and maintaining class attributes. Furthermore, RFE optimizes the model by methodically picking the most relevant features, enhancing interpretability and performance. The distinctive integration of sophisticated class balance and optimum feature selection distinguishes XRFILR, making it a more resilient and precise method for early-stage Parkinson's disease prediction. The methodology section outlined an (XRFILR) for predicting Parkinson's disease using speech data and a logistic regression-based explainable feature importance model. It involves data preprocessing, feature selection, machine learning models, and explainable AI techniques. The hybrid technique improves accuracy and interpretability by focusing on critical features and addressing dataset imbalances, enhancing the overall model's performance.

## IV. RESULTS ANALYSIS AND DISCUSSION

This part delineates and analyzes the experimental data, focusing on feature extraction, feature selection, oversampling, and model assessment outcomes. We employed Recursive Feature Elimination with Logistic Regression for feature extraction and utilized eXplainable Artificial Intelligence for model interpretability. The PD voice dataset utilizes diverse classification techniques, and its efficacy has been evaluated. Compared to ML models, the proposed model (XRFILR) achieves the highest testing accuracy (96.46%) while maintaining a balance between precision, recall, and F1 score. RFE-based feature selection works incredibly well with this dataset when combined with logistic regression. Models that perform better with XGBoost achieve a testing accuracy of 95. Because of its ability to handle unbalanced data effectively, the proposed model is a strong contender for the classification of Parkinson's disease. The MLP Classifier also performed admirably, achieving 94.25% on all measures. Overfitting issues random forest, MLP, and CatBoost models show perfect training accuracy but a drop in testing accuracy, which may indicate overfitting. This emphasizes how vital feature selection techniques like RFE are for reducing overfitting. With an accuracy of 90.71%, KNN performs the worst out of all the models, making it less suitable for this classification task given the better results from other models shown in Table 3. Figure 6 shows the model performance comparison based on training, testing, precision, recall, and f1 score.

The juxtaposition of the two tables, one employing KMeansSMOTE and PCA and the other utilizing RFE with Logistic Regression, demonstrates significant enhancements in model efficacy for predicting PD from speech data. All models exhibit uniform enhancement in testing accuracy, precision, recall, and F1 scores upon using RFE with logistic regression. The SVC demonstrates notable enhancement, increasing testing accuracy from 89.38% to 93.36%. Likewise, Random Forest, MLP, XGBoost, CatBoost, and K-Nearest Neighbors (KNN) classifiers perform better when employing the RFE methodology. The XGBoost classifier attains a testing accuracy of 95.13% and an F1 score of 95.12%, ranking among the top in Table 4.

The proposed model's superiority is underscored by the comparative performance of various machine learning models that employ various feature selection techniques, as illustrated in Table 4. This investigation implemented an ablation-type approach to assess various feature selection methodologies, including RFE with Logistic Regression and PCA with Logistic Regression. The results indicate that the proposed model, RFE with Logistic Regression, obtained a significantly higher testing accuracy of 96.46% than PCA with Logistic Regression, which attained 87.17%. The ablation analysis emphasizes the proposed approach's resilience and efficacy, which isolates the effects of individual components. Table 4 also demonstrates that the MLP Classifier and XGBoost Classifier obtained high training accuracies; however, their testing accuracies (90.27% and 89.38%, respectively) were inferior to those of the proposed model. Likewise, Table 3 underscores the proposed model's superior performance and robustness compared to other classifiers and feature selection methods, as evidenced by its superior precision, recall, and F1 scores for speech-based Parkinson's Disease diagnosis.

The comparison indicates that RFE with Logistic Regression substantially enhances all machine learning models. The XGBoost and proposed classifier (RFE with Logistic Regression) in table 3 demonstrate superior performance, with XGBoost attaining an accuracy of 95.13% and the proposed classifier exceeding all others with a testing accuracy of 96.46%, alongside balanced precision, recall, and F1 scores. Overall, the models exhibit improved performance with RFE, which presumably optimizes selecting the most pertinent features, resulting in enhanced classification efficacy. This indicates that feature selection by RFE with logistic regression improves model efficacy for PD prediction, with the proposed model emerging as the most dependable model for this task, according to these findings.

The model accurately classified 111 cases as positive and 107 as unfavorable, with a high precision and recall of 96.5%. It also identified four cases as positive but misclassified as negative. The model's F1-score of 0.965 reflects its performance, as shown in Figure 7. It was trained in speech, demonstrating its effectiveness in speech-related outcomes predicting. The model's efficacy is noteworthy.

The most reliable models for speech data are Logistic Regression and MLP Classifier, as they attain the maximum AUC of 0.99, as shown in Figure 8. Random Forest, SVM, CatBoost, and XGBoost exhibit exceptional performances, with AUC values 0.98. These models are commendable; nonetheless, they are slightly less effective than the leading performers. KNN still achieves an AUC of 0.97 but encounters slight difficulties relative to the other models. This is anticipated, as KNN may struggle to handle the intricacies of high-dimensional voice data successfully. The AUC values for models including MLP, Random Forest, and XGBoost are high; nevertheless, as prior performance measurements suggest, additional fine-tuning may be necessary to mitigate overfitting. The comparison of ROC curves highlights the effectiveness of several models in detecting Parkinson's disease through speech-based data. The MLP Classifier and Logistic Regression are the most efficacious models, exhibiting AUC values of 0.99, rendering them the most appropriate choices for this undertaking. Random Forest, SVM, XGBoost, and CatBoost exhibit comparable efficacy since their roughly aligned performances. The strong predictive indicators offered by the speech-based feature set employed in this study render it a promising method for the early identification of Parkinson's disease. Future enhancements may include adjusting hyperparameters in models like Random Forest and XGBoost or exploring ensemble approaches to augment forecast accuracy.

The SHAP summary plot demonstrates the influence of diverse speech-based features on the prediction of Parkinson's disease, utilizing a machine learning model trained. The horizontal axis denotes SHAP values, which measure the extent to which a particular factor affects the model's output, with positive values enhancing the probability of a Parkinson's diagnosis and negative values diminishing it. Each row represents a feature ranked by its significance, with the foremost features (such as Feature 41, Feature 25, and Feature 26) exerting the most significant influence on the model's predictions. The color gradient indicates the value of each feature, with red representing higher values and blue indicating lower values, demonstrating the influence of each feature's magnitude on predictions.

Feature 41 indicates that elevated values enhance the probability of Parkinson's disease, whereas diminished values decrease it, implying that specific speech alterations

**TABLE 3. Performance comparison of ML techniques in (XRFILR) model.**

| ML Model | Training Accuracy (%) | Testing Accuracy (%) | Precision Score (%) | Recall Score (%) | F1 Score (%) |
|---|---|---|---|---|---|
| Support Vector Classifier | 97.35 | 93.36 | 93.93 | 93.36 | 93.33 |
| Random Forest Classifier | 100.00 | 92.48 | 93.03 | 92.48 | 92.45 |
| MLP Classifier | 100.00 | 94.25 | 94.25 | 94.25 | 94.25 |
| XGBoost Classifier | 100.00 | 95.13 | 95.41 | 95.13 | 95.12 |
| CatBoost Classifier | 100.00 | 92.92 | 93.13 | 92.92 | 92.91 |
| KNN Classifier | 92.37 | 90.71 | 90.71 | 90.71 | 90.71 |
| Proposed Classifier (RFE with Logistic Regression) | 97.23 | 96.46 | 96.46 | 96.46 | 96.46 |

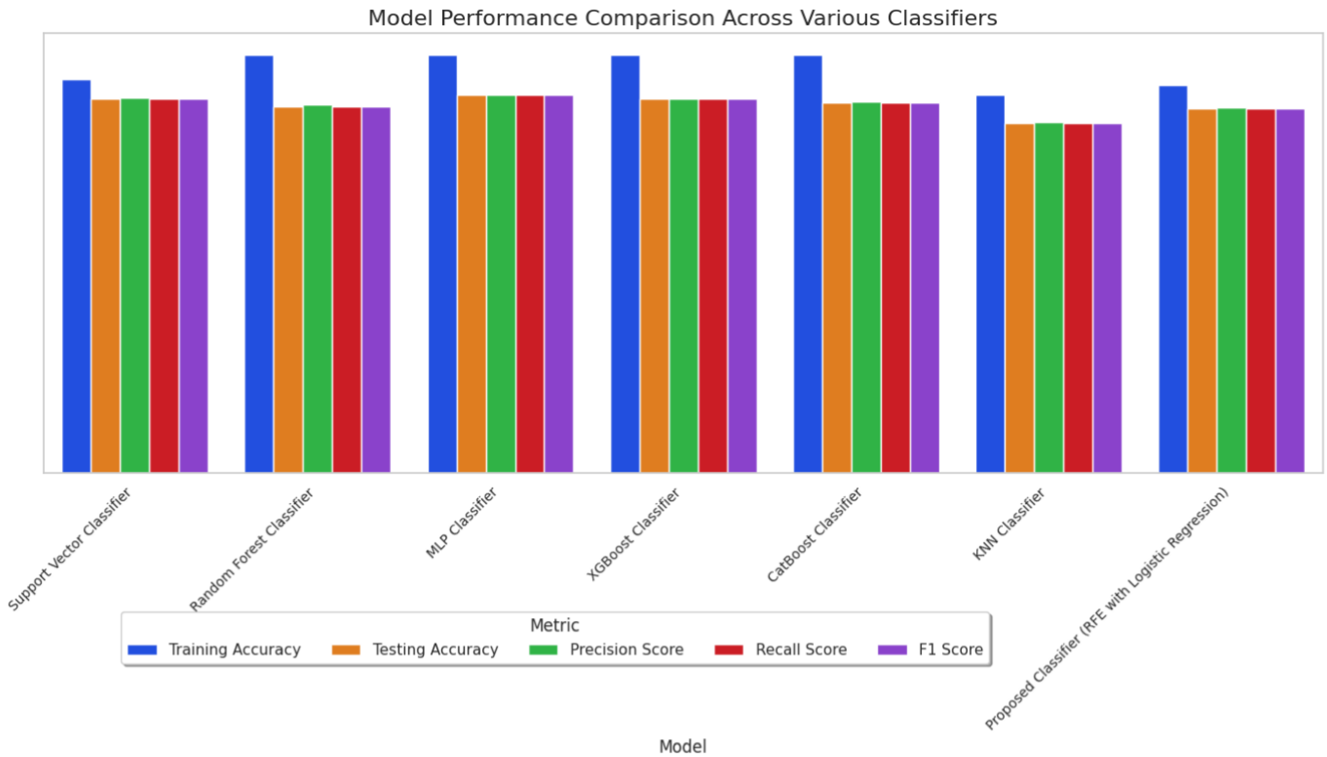



**FIGURE 6. Model performance comparison across various classifiers.**

**TABLE 4. Performance comparison of ML techniques in KMeansSMOTE and PCA.**

| ML Model | Training Accuracy (%) | Testing Accuracy (%) | Precision Score (%) | Recall Score (%) | F1 Score (%) |
|---|---|---|---|---|---|
| Support Vector Classifier | 94.02 | 89.38 | 89.65 | 89.38 | 89.34 |
| Random Forest Classifier | 100.00 | 87.61 | 88.13 | 87.61 | 87.53 |
| MLP Classifier | 100.00 | 90.27 | 90.29 | 90.27 | 90.27 |
| XGBoost Classifier | 100.00 | 89.38 | 89.41 | 89.38 | 89.37 |
| CatBoost Classifier | 100.00 | 88.50 | 88.65 | 88.50 | 88.46 |
| KNN Classifier | 90.37 | 83.63 | 83.80 | 83.63 | 83.64 |
| Proposed Classifier (PCA with Logistic Regression) | 92.58 | 87.17 | 87.21 | 87.17 | 87.17 |

represented by this feature correlate with PD symptoms. Conversely, feature 25 exhibits an inverse correlation, wherein diminished values enhance the model's prediction of Parkinson's, underscoring that a reduction in specific vocal capabilities may signify the condition. Figure 9 demonstrates non-linear behavior in certain features, where high and low values influence predictions differently, signifying intricate relationships within the data. The model's sensitivity to variations in many features ensures the reliable identification of modest speech defects associated with Parkinson's disease. This visual depiction elucidates how specific speech qualities influence the model's judgments, augmenting the

**TABLE 5.** Comparison of the proposed work with existing works for PD prediction.

| Study | Speech Dataset | Preprocessing Method | Feature Selection Method | Explainable AI | Classification Method | Accuracy (%) |
|---|---|---|---|---|---|---|
| Hawi et al., 2022 [23] | UCI ML Repository | Normalization and Standardization | MFCC (Mel frequency cepstral coefficient) | No | Random Forest | 88.84 |
| Rohit Lamba et al., 2022 [22] | UCI ML Repository | No | MIRFE | No | Random Forest and XGBoost Classifier | 93.88 |
| Mithun Shivakoti et al., 2023 [41] | UCI ML Repository | Random Oversampling | No | No | CatBoost | 95.1 |
| Nutan Singh et al., 2024 [30] | UCI ML Repository | SMOTE | Ensemble Classifier | No | EFSA Algorithm | 90.2 |
| Syed Nisar H B. et al., 2024 [12] | UCI ML Repository | SMOTE | PCA | No | AdaBoost | 96 |
| Proposed Work XRFILR Model | UCI ML Repository | KMEANSSMOTE | RFE with LR | SHAP | LR Classifier | 96.46 |

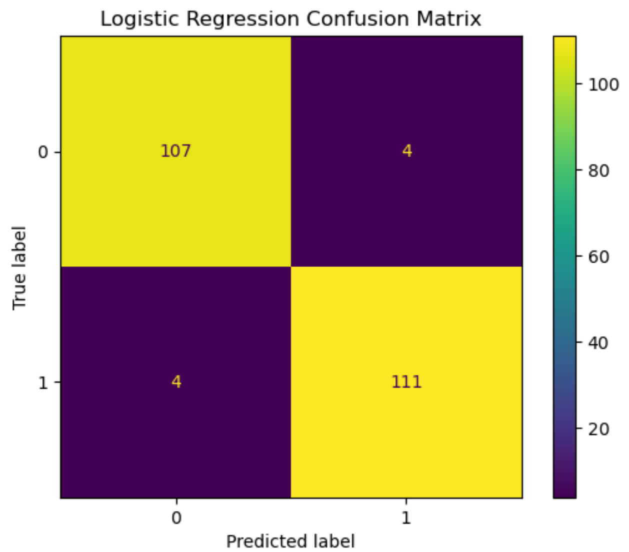



**FIGURE 7.** Confusion matrix predicting RFE with logistic regression.

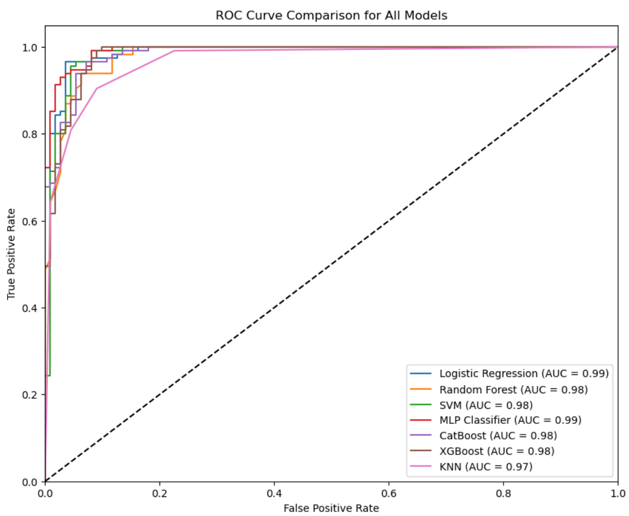



**FIGURE 8.** Comparison of ROC curves for various machine learning algorithms.

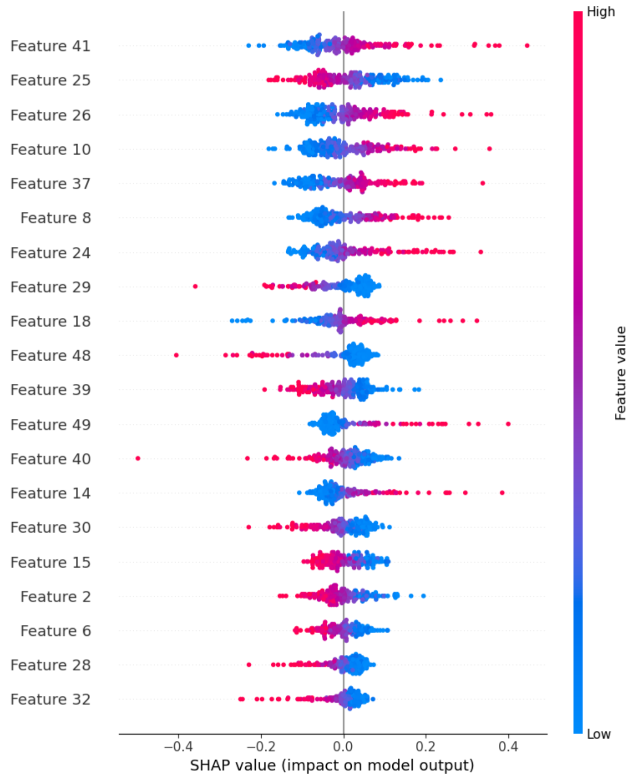



**FIGURE 9.** Feature importance analysis with SHAP.

interpretability and reliability of AI-driven diagnostics. These findings assist physicians by identifying essential vocal markers and facilitate the creation of individualized diagnostic tools, rendering explainable AI a significant resource for the early identification and non-invasive monitoring of Parkinson's disease.

### A. COMPREHENSIVE COMPARATIVE ANALYSIS

The proposed model significantly improves previous related works for speech-based Parkinson's disease prediction. While all research studies use the UCI ML Repository dataset, the proposed work combines KMeansSMOTE for sophisticated data balancing and Recursive Feature Elimination (RFE) with Logistic Regression (LR) for robust feature selection. Unlike existing research models that lack interpretability, the suggested model contains SHAP (SHapley Additive exPlanations), allowing for explainable AI, which is critical for trust and openness in medical applications. Despite adopting a more straightforward classification approach (Logistic Regression), the suggested model achieves 96.4% accuracy, outperforming most previous works and matching the performance of more complicated classifiers such as AdaBoost. This combination of high accuracy, effective preprocessing, and explainability distinguishes the proposed model as a better and more comprehensive solution than previous methods.

The comparative analysis indicates that the proposed XRFILR model for predicting Parkinson's Disease surpasses other current methodologies in accuracy and model interpretability. Although several research studies attain excellent accuracy with classifiers such as SVM, CNN, LSTM, and deep neural networks, they frequently exhibit deficiencies in robust feature selection techniques and explainability. The XRFILR model attains an accuracy of 96.46% and incorporates KMeans SMOTE for addressing class imbalance and SHAP for enhanced interpretability in artificial intelligence.

This makes the proposed model a more dependable and thorough solution for PD prediction compared to current methodologies. Integrating sophisticated preprocessing, feature selection, and explainability offers a considerable benefit for clinical applications, augmenting confidence and usefulness in practical situations. Table 5 delineates much research on the classification of Parkinson's Disease (PD) utilizing voice datasets, highlighting variations in preprocessing, feature selection, classification methods, and model accuracy. The existing related works employ the UCI Machine Learning Repository, including preprocessing techniques such as normalization, standardization, random oversampling, and SMOTE to address class imbalance. Feature selection strategies differ, encompassing methods like MFCC and PCA. However, several research omits this procedure. Classification methodologies encompass ensemble techniques like Random Forest, XGBoost, and AdaBoost. The proposed model integrates KMeansSMOTE for data balancing, Recursive Feature Elimination (RFE) for feature selection, and SHAP for interpretability, employing a Logistic Regression classifier. This model attains a peak accuracy of 96.46%, demonstrating that a comprehensive strategy involving efficient preprocessing, feature selection, and explainable AI may improve PD classification precision. The proposed model has distinct properties that differentiate it from existing methodologies. KMeansSMOTE is used for advanced data balancing, improving the representation of minority classes while preserving feature distribution. Integrating Recursive Feature Elimination (RFE) with Logistic Regression enhances feature selection by preserving the most critical features, improving model performance and interpretability. Furthermore, SHAP (SHapley Additive exPlanations) improves explainability, filling a significant gap in most previous solutions and making the model more transparent and trustworthy for clinical applications. The model achieves a high accuracy of 96.46%, beating or matching other cutting-edge approaches, all while using a more straightforward categorization methodology. However, the model does have certain limitations. The preprocessing procedures, notably KMeansSMOTE and RFE, add computational complexity and may need more processing time than straightforward approaches such as regular SMOTE. Furthermore, although Logistic Regression assures interpretability, it may not be as good at capturing complicated non-linear patterns as sophisticated classifiers such as Random Forest or AdaBoost. Despite these restrictions, the model's benefits in accuracy, openness, and feature choices make it a solid and dependable option. The model's superior accuracy and transparency are shown, for example, when compared to experiments conducted by Rohit Lamba et al. (93.88%), Syed Nisar et al. (96%), and Hawi et al. (88.14%). In a fair assessment, this comparison will show how well the proposed model holds up and how it adds anything new to the field.

### B. DISCUSSION

The research and experimental findings unequivocally indicate that the proposed XRFILR model (Recursive Feature Elimination with Logistic Regression) for predicting Parkinson's Disease (PD) provides substantial benefits compared to current machine learning techniques. The performance data demonstrate that the model attains a peak testing accuracy of 96.46% while sustaining balanced precision, recall, and F1 score, surpassing competing models, including SVM, Random Forest, MLP, and XGBoost. The efficacy of the proposed model may be ascribed to several variables, particularly the implementation of Recursive Feature Elimination (RFE) for feature selection and KMeansSMOTE for addressing class imbalance in the dataset. RFE is essential to this model's exceptional performance, facilitating the selection of the most pertinent features and directly influencing the model's capacity to generalize effectively to unknown data. Integrating RFE with logistic regression significantly mitigates overfitting, a problem prevalent in classifiers such as Random Forest, MLP, and CatBoost, which attain flawless training accuracy but significantly decline testing accuracy, signifying overfitting. The suggested model utilizes SHAP (SHapley Additive exPlanations) to boost explainability. Increasing the model's transparency is essential for clinical applications where comprehending feature significance is crucial for decision-making.

The comparison of models utilizing KMeansSMOTE and PCA with those employing RFE together with logistic regression demonstrates the superiority of RFE in improving model performance. Specifically, classifiers like SVC, Random Forest, and XGBoost exhibit significant enhancements when integrated with RFE, affirming its efficacy in optimizing feature selection. Although XGBoost and other models demonstrate good performance, the suggested XRFILR model is distinguished by its exceptional balance across all criteria, rendering it the most dependable choice for PD prediction. The SHAP summary graphic offers further insights into the impact of specific features from the speech dataset on the model's judgments, enhancing its overall robustness. The capacity to discern nuanced speech differences linked to PD renders the model exceptionally appropriate for early identification, a paramount objective in PD diagnosis.KMeansSMOTE improves the model's ability to tackle class imbalance in the dataset, ensuring enough representation of the minority class (healthy individuals) during training.

In summary, the proposed XRFILR model exhibits its efficacy in harmonizing predictive accuracy, feature selection, and interpretability, rendering it an optimal alternative for predicting PD using speech data. Compared to alternative models, it proficiently tackles issues such as class imbalance, overfitting, and insufficient transparency. The model's superior performance across several metrics such as accuracy, precision, recall, and F1 score, coupled with its implementation of explainable AI (SHAP), highlights its applicability in real-world clinical settings. It delivers a thorough solution that accurately predicts PD while delivering interpretable data, increasing confidence in AI-driven diagnostic tools. This model is a formidable candidate for future endeavors in Parkinson's disease diagnosis, providing significant insights for medical practitioners and researchers alike.

## V. CONCLUSION AND FUTURE WORK

The XRFILR model proposed in this work incorporates essential phases such as data preparation, exploratory analysis, class balance, feature selection, and explainable AI to predict Parkinson's Disease (PD) utilizing speech data. The model utilizes the KMeansSMOTE approach to rectify the class imbalance in the dataset, facilitating more equitable predictions. Recursive Feature Elimination (RFE) with Logistic Regression optimizes model performance by identifying the most relevant features from the speech dataset, decreasing dimensionality and enhancing computing efficiency. Using Explainable Artificial Intelligence (XAI), particularly SHAP values, offers significant insights into feature significance, enhancing interpretability and confidence in the model's predictions. The model's efficacy was assessed utilizing machine learning classifiers, such as Logistic Regression, Random Forest, SVM, MLP, CatBoost, XGBoost, and KNN, on a dataset divided into training and testing subsets. Metrics including accuracy, precision, recall, F1-score, and ROC-AUC were employed to evaluate the model's efficacy in classifying speech samples as either Parkinson's Disease or healthy control. The proposed model has significant predictive potential, producing precise findings and providing a solid foundation for early Parkinson's disease identification using non-invasive speech analysis.

Future research could expand the XRFILR model to tackle complex classification challenges, including multi-class and multi-label issues, predicting Parkinson's Disease phases. This could be achieved by integrating advanced preprocessing methods, optimizing features, using multimodal data, real-time data gathering, deep learning methodologies, Explainable AI, and cross-institutional cooperation. These advancements could improve model efficacy, generalizability, and resilience across diverse populations.

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